Properties of Eigen values

→ Consider $A_{n\times n}$ and $\lambda_1, \lambda_2 \ldots \lambda_n$ are its eigen values

→ Sum of E.values = Trace(A)

→ Product of E-values = $|A|$

→ 0 is an E.values of A iff A is singular

→ No. of non-zero E-values of A $\leq \rho(A)$

→ E-values of UTM, LTM, diagonal matrix scalar matrix are just the diagonal element.

→ Don't apply E-operations in a given matrix while calculating E.values becoz equivalent matrices have different E-values.

→ E-values ~~are~~ of A^T are same i.e. $\lambda_1, \lambda_2 \ldots \lambda_n$

→ E.values of A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m \ldots \lambda_n^m$

→ E.values of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2} \ldots \frac{1}{\lambda_n}$ $\because [|A| \neq 0]$

→ E.values of (adj A) are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \ldots \frac{|A|}{\lambda_n}$

→ If sum of all the elements in each column or each row is unique constant k then k will be the one of the E.values of that matrix.

→ If λ is an E.values of matrix A then to find the E.values of any algebric expression formed by matrix A can be obtained by replacing $A \to \lambda$ in that expression.